

XAI



EASA COLLEGE
OF ENGINEERING & TECHNOLOGY (ECET)

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XAI-Based Intelligent Disease Prediction System with Transparent Decision Insights

Project Guide:

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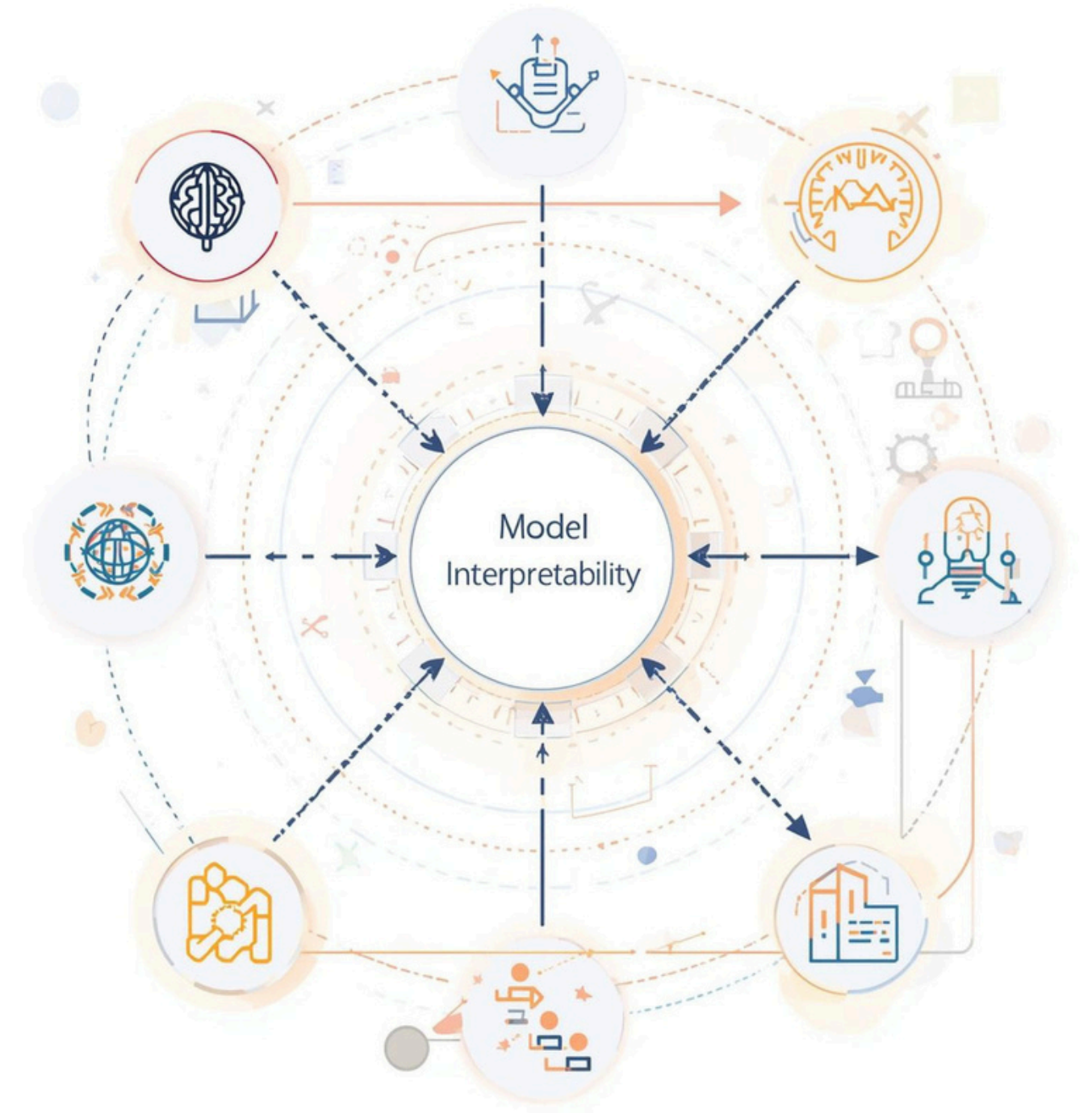
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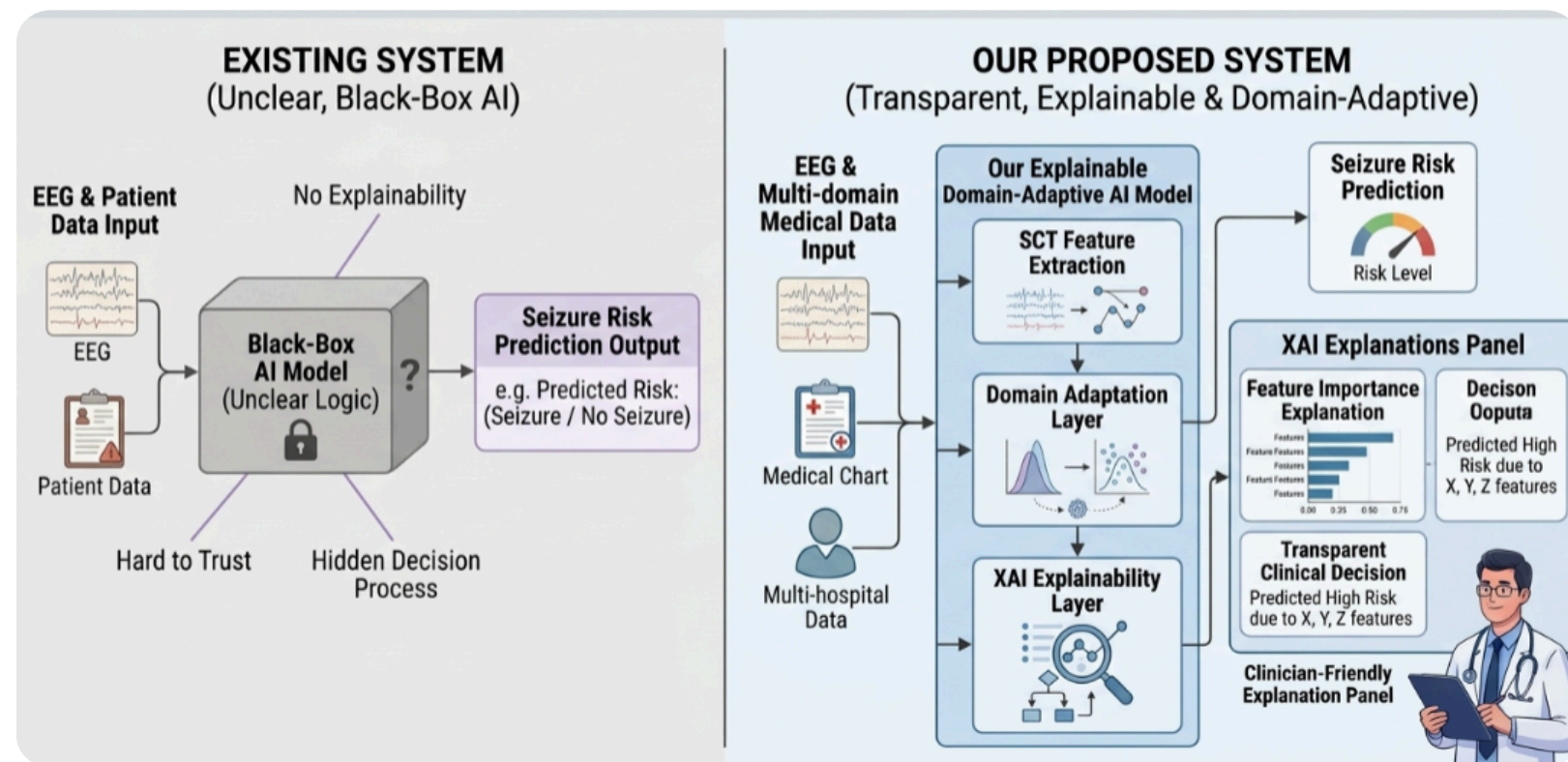


Introduction to XAI

- **Epilepsy are serious neurological disorders requiring early detection.**
- **Traditional diagnosis using EEG and clinical reports is time-consuming and expert-based.**
- **AI and Machine Learning enable faster and more accurate disease prediction.**
- **However, most ML models act as black-box systems without clear explanations.**
- **This lack of interpretability reduces trust in clinical decision-making.**
- **Explainable AI (XAI) provides transparent insights into model predictions.**
- **Our project combines ML classification with XAI techniques for reliable and interpretable disease prediction.**



PROBLEM STATEMENT



In **current medical AI systems**, models may give accurate results but often fail to explain their reasoning. For diseases like **Epileptic Seizure**, doctors need to know why a patient is **classified as high-risk** — not just the final output.

Most **existing systems**:

- **Don't provide clear feature-based** explanations (which signals or factors caused the prediction).
- Are **built on complex deep learning** architectures that are hard to interpret.
- **Lack a proper interface** that medical professionals can easily use.

So, the **problem is to design a system** that is **both intelligent and transparent** — one that maintains **AI accuracy** while making its decision-making process **understandable to humans**.

System Architecture

- **EEG Data Input**

Multi-channel EEG signals (CHB-MIT dataset, 18 channels, 256 Hz)

- **Signal Preprocessing**

Bandpass & notch filtering, artifact removal, sliding window segmentation, pre-ictal & inter-ictal labeling

- **Feature Generation**

Raw EEG (time-domain), Spectrogram (time-frequency), Spike Encoding (neuromorphic representation)

Model Layer

SCT (spatio-temporal learning), GRL (domain adaptation), DA-SCT (hybrid model), Federated Learning (privacy-preserving training)

- **Prediction**

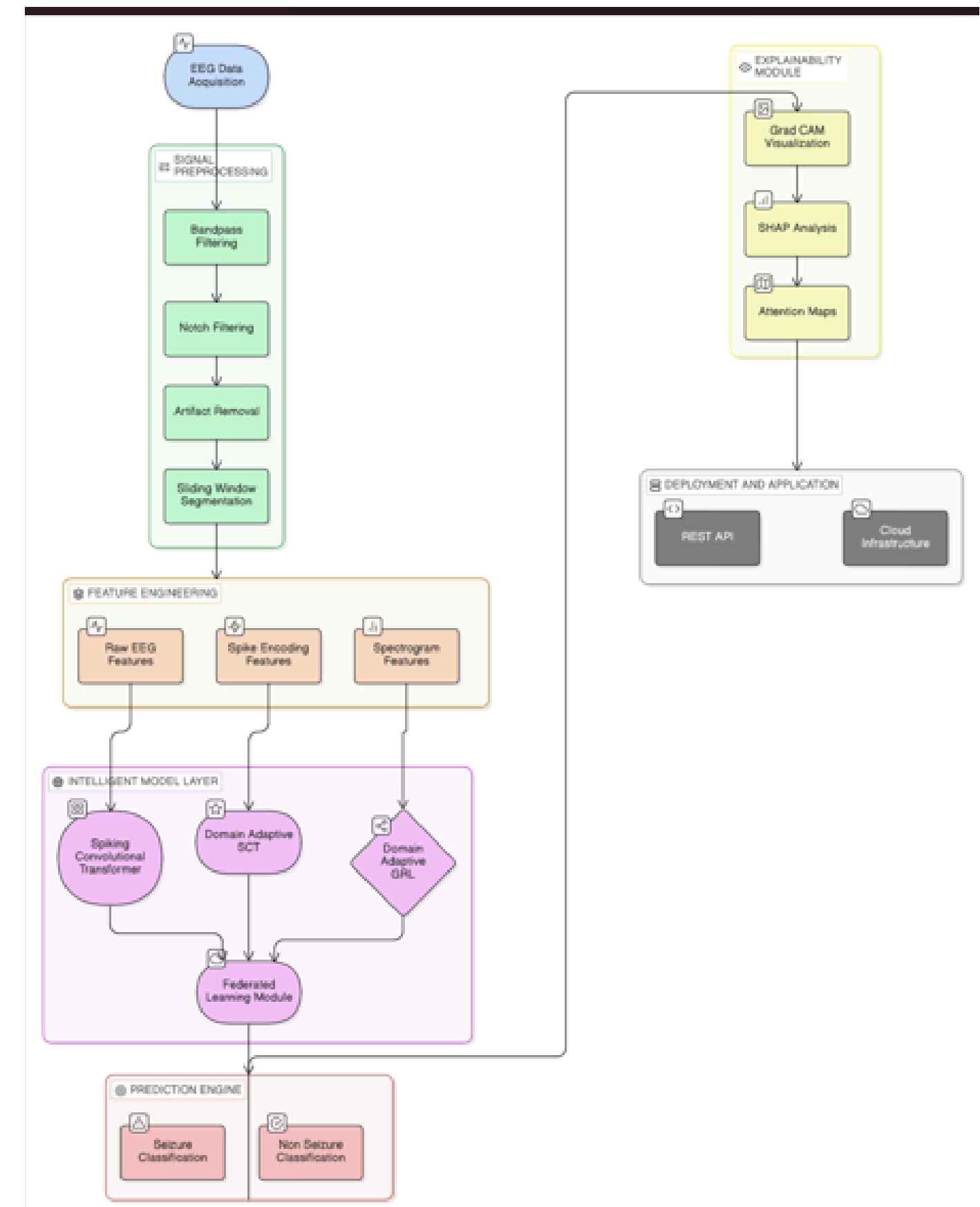
Seizure vs Normal classification with probability score and early warning **capability**

- **Explainable AI (XAI)**

Grad-CAM (region importance), SHAP (feature contribution), Attention maps (model focus visualization)

- **Deployment**

FastAPI (real-time inference API), Streamlit (interactive dashboard), Cloud deployment (scalable access)





Literature Survey

S.No	Author & Year	Title of the Paper	Publication	Dataset	Findings / Contribution
1	Md. Zakir Hossain Zamil et al. (2025)	Stroke Prediction on Healthcare Data Using SMOTE and Explainable Machine Learning	IEEE International Symposium on Digital Forensics and Security	Healthcare Dataset (5,110 records)	Applied SMOTE with RandomForest, XGBoost, and CatBoost. CatBoost achieved 95.58% accuracy , and SHAP analysis identified top stroke risk features
2	Ruifeng Zheng et al. (2025)	EEG-based Epileptic Prediction via a Two-Stage Channel-Aware Set	arXiv Preprint	CHB-MIT EEG Dataset	Proposed two-stage transformer model reducing EEG sensors from 18 to 2.8 while maintaining 80.1% sensitivity and 95.5% specificity .
3	Heewon Kim et al. (2024)	Learning with Noisy Labels: Interconnection of Two Expectation-Maximizations	arXiv Preprint	Image & Biomedical Data	Introduced a dual EM-cycle learning approach to handle noisy labels in deep learning; improved classification stability and accuracy.

Literature Survey

S.No	Author & Year	Title of the Paper	Publication	Dataset	Findings / Contribution
4	Zahra Sadeghi et al., 2024	A Review of Explainable Artificial Intelligence in Healthcare	Elsevier – <i>Computers & Electrical Engineering</i>	Multiple Medical Datasets	Reviewed modern XAI techniques (SHAP, LIME, Grad-CAM) and their use in clinical AI systems for trust, interpretability, and accountability in decision support.
5	Simone Sarti et al., 2023	Enhancing Once-For-All: A Study on Parallel Blocks, Skip Connections and Early Exits	arXiv Preprint (2302.01888v1)	Tiny ImageNet Dataset	Proposed OFAv2 architecture adding dense skip connections and early exits to Once-For-All networks, improving accuracy by 12.07 % and energy efficiency for deep learning tasks.

Summary of Literature Survey

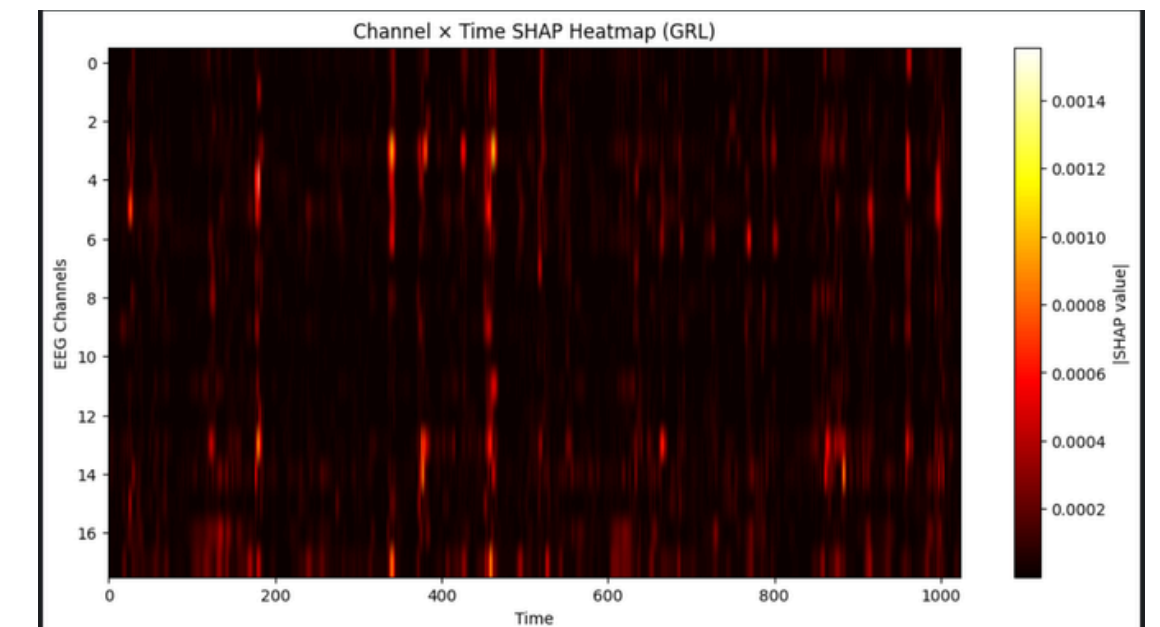
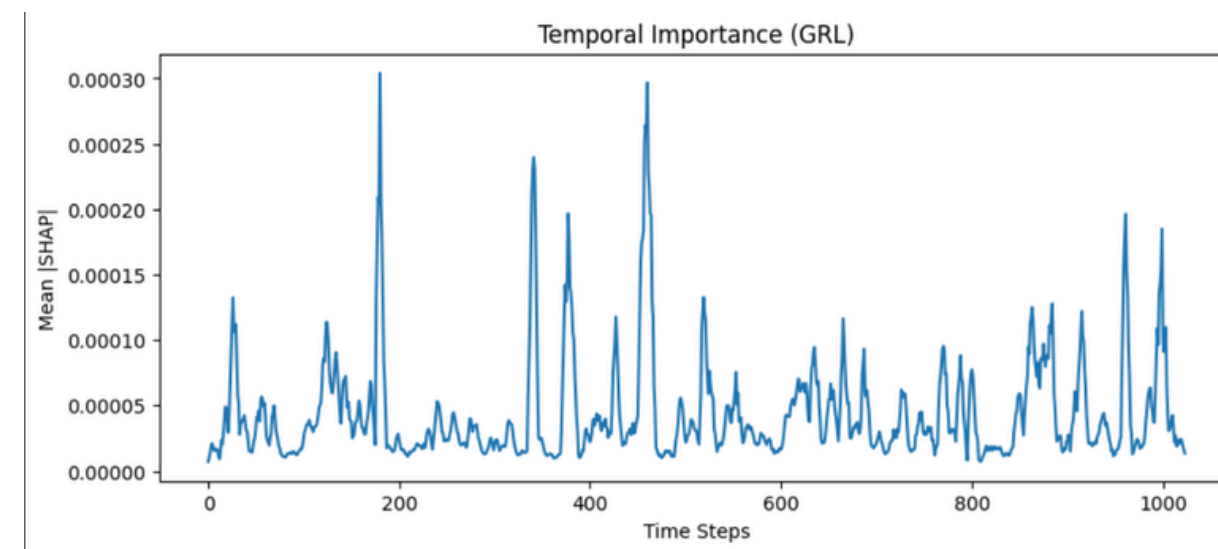
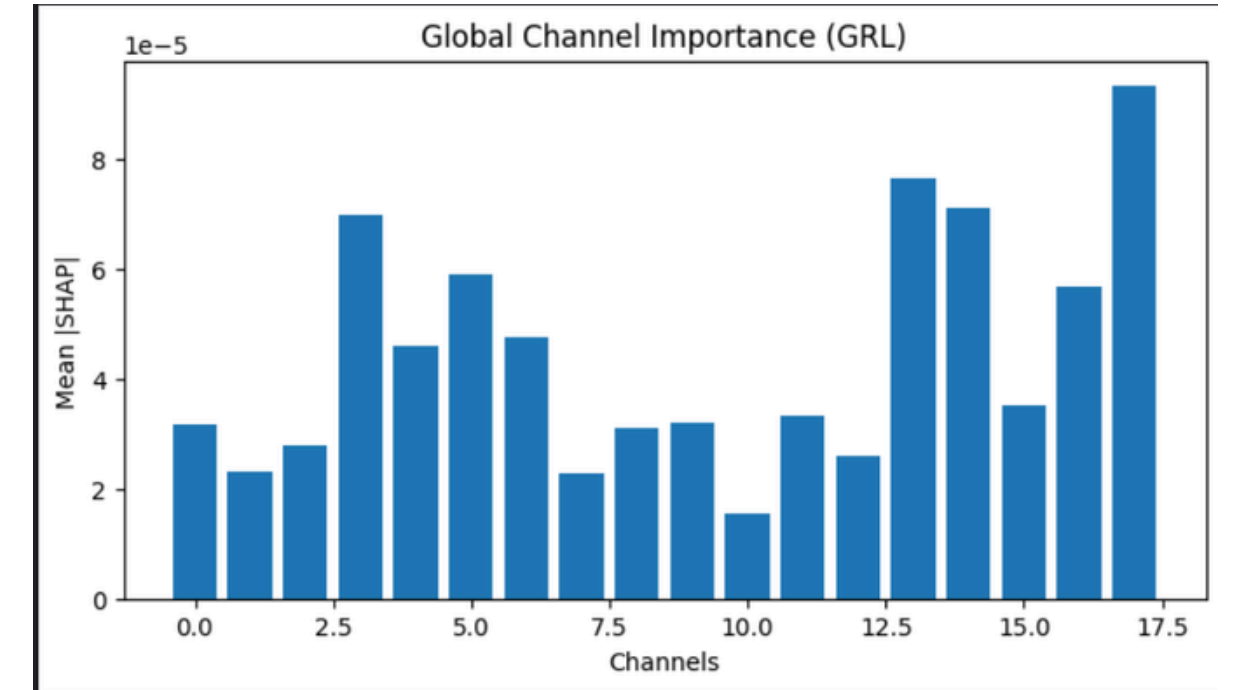
Title	Stroke Prediction on Healthcare Data Using SMOTE and Explainable Machine Learning	
Author Name	Ruifeng Zheng et al. (2025)	Arun Pandian et al. (2026)
Model	LSTM, CNN, Basic ML models	- Long Short Term Memory(LSTM-CNN) - Spiking Convolutional Transformer (SCT) - Gradient Reversal Layer Domain Adaptation Network (GRL-DAN) - Domain-Adaptive Spiking Conv Transformer (DA-SCT) - Privacy-Preserved Federated Learning (FedAvg + Differential Privacy)
Dataset	PhysioNet CHB-MIT Pediatric EEG Dataset (23 patients, ~900+ hours EEG, 198 seizures)	PhysioNet CHB-MIT Pediatric EEG Dataset (23 patients, ~900+ hours EEG, 198 seizures)
Accuracy	97.08%	
Explainability	Not aplicable	SHAP feature interpretation for seizure risk
Year	2025 : IEEE : APRIL	

Project Objectives {X-AI}

The goal of this project is to create an Explainable AI-based Disease Prediction System that focuses on Stroke and Epilepsy detection.

We aim to:

1. Build accurate prediction models using RandomForestClassifier and Spiking Convolutional Transformer.
2. Apply Explainable AI tools like SHAP and LIME to show how each input feature affects the final decision.
3. Visualize decision insights so doctors can interpret AI outputs clearly.
4. Develop a user-friendly dashboard for clinical use.
5. Ensure model transparency to promote medical trust and real-world usability.



Project Timeline

Baseline Model Training

RandomForest baseline model trained and initial performance evaluated.

14.02.2026 – I Review

SCT Model Development

SCT model built and compared with baseline for improved prediction.

21.04.2026 – III Review

Data Collection & Preprocessing

EEG datasets collected, cleaned, and baseline model prepared.

22.01.2026 – 0th Review

XAI Explainability Integration

SHAP and LIME applied to explain model predictions and validate relevance.

04.04.2026 – II Review

Final Documentation & Submission

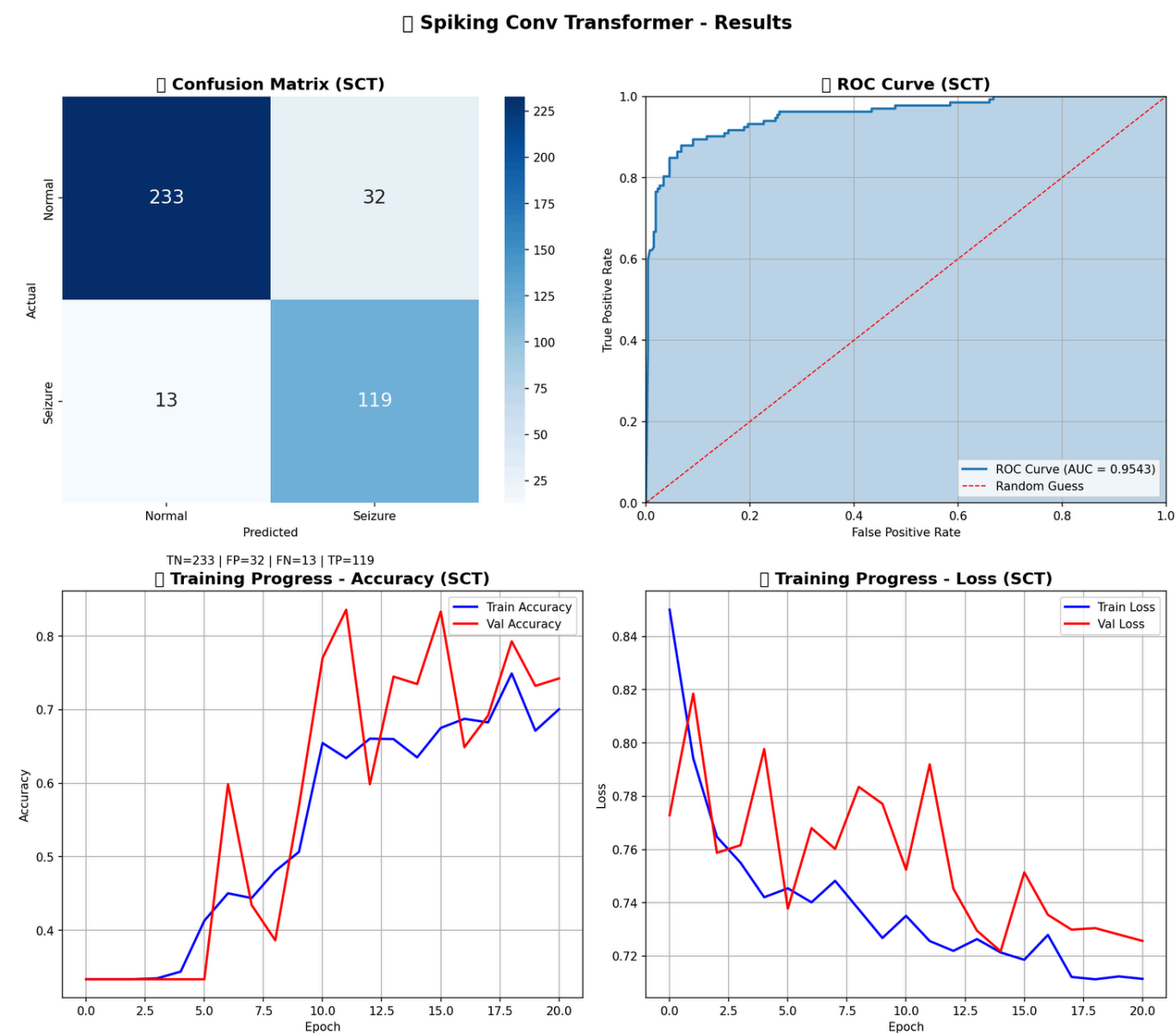
Project report and presentation finalized and submitted.

29.04.2026 – Project Submission

Current Implementation



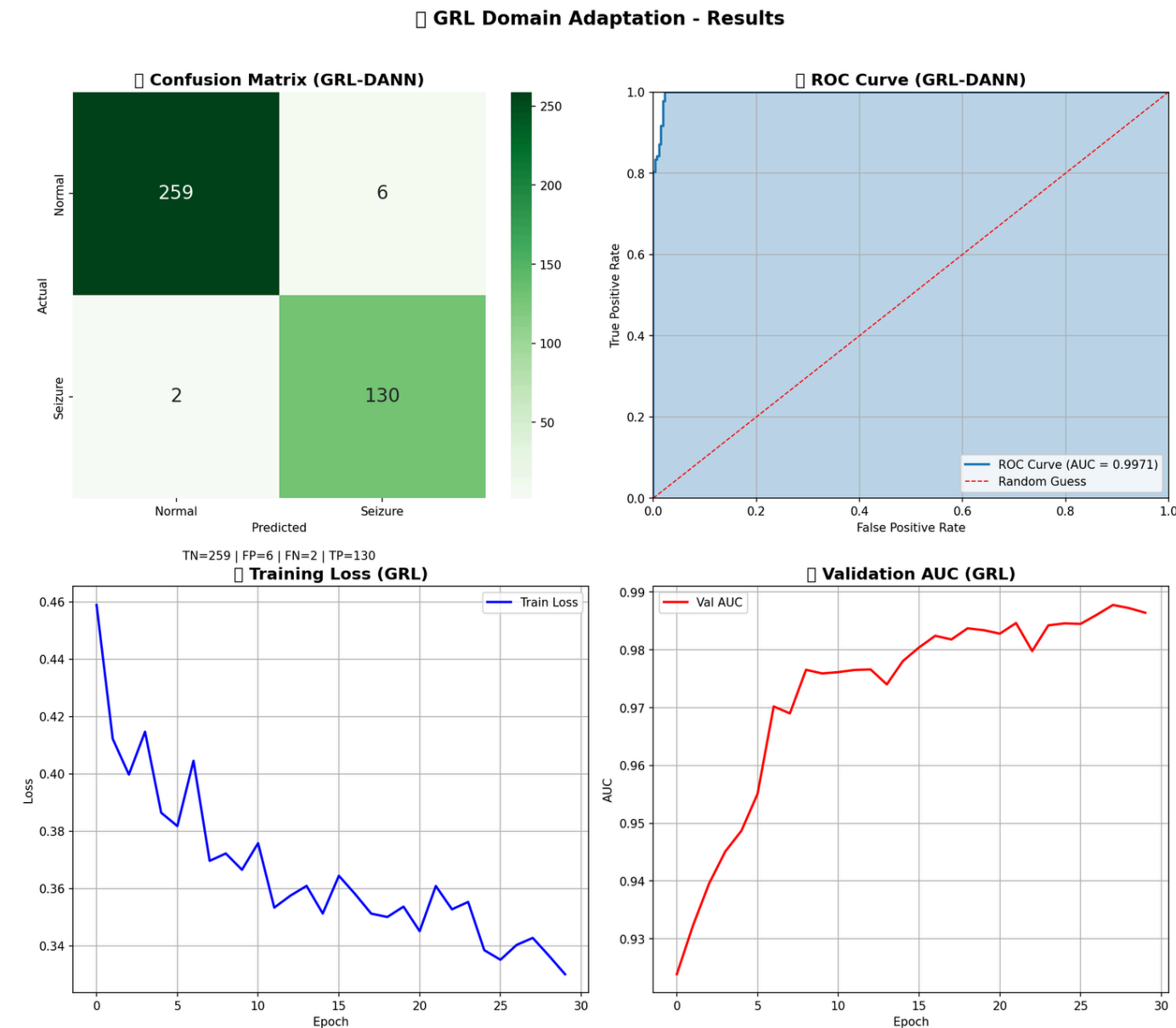
(1)



(2)

At this stage, we have collected and prepared **EEG-based** medical datasets from trusted sources such as Kaggle and PhysioNet, focusing on Stroke and Epilepsy conditions. A **SCT AND GRL-DAN** has been implemented as the baseline model. The data is normalized using StandardScaler and split into **60% for training and 30% for validation and 30% for testing**. The model has been successfully trained and evaluated, showing promising accuracy and reliable prediction performance for today's demonstration.

GRL (GRADIENT REVERSAL LAYER)



The GRL-DANN model achieves highly reliable seizure prediction with strong cross-patient generalization. The confusion matrix shows very few **false positives (6)** and **false negatives (2)**, indicating high sensitivity and **low false alarm rate**. The ROC curve reaches an **AUC of 0.9971**, demonstrating near-perfect class separation. Training loss decreases steadily, and validation AUC improves to around **0.99**, confirming stable learning and effective domain adaptation. Overall, GRL provides robust and clinically reliable seizure detection.

Model Comparision I

Model Comparison (4 Models)

Model	Accuracy	Recall	F1	AUC
CNN-LSTM	0.97	0.99	0.96	1.00
SCT	0.90	0.79	0.85	0.95
GRL	0.97	0.96	0.96	0.99
DA-SCT	0.96	0.93	0.94	0.99

This table compares four deep learning models:

- CNN-LSTM
- Spiking Convolutional Transformer (SCT)
- Gradient Reversal Layer Network (GRL)
- Domain Adaptation Spiking Conv Transformer (DA-SCT)

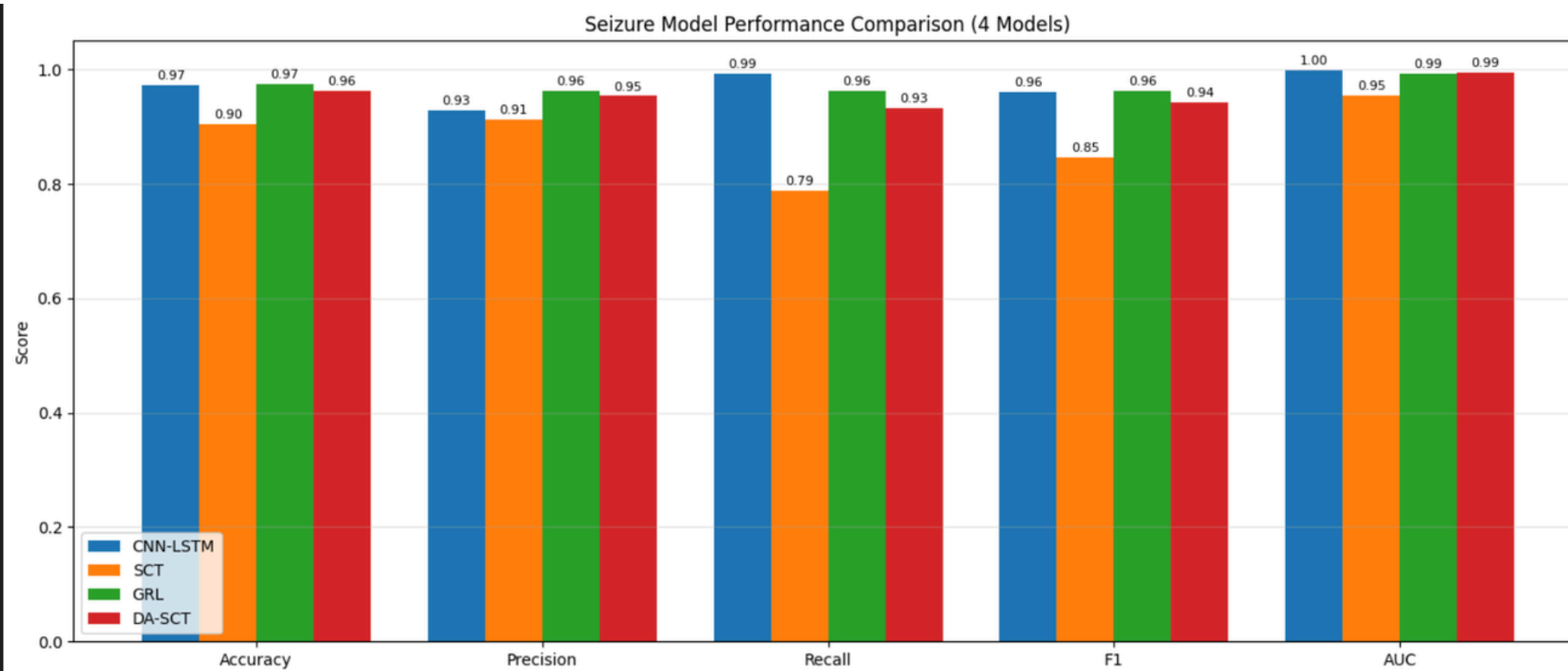
Evaluation Metrics Used:

- Accuracy
- Recall (Sensitivity)
- F1 Score
- AUC (Area Under ROC Curve)

Purpose:

To analyze detection capability, generalization performance, and seizure sensitivity across models.

Model Comparison II



This graph compares four seizure prediction models across Accuracy, Precision, Recall, F1-score, and AUC.

CNN-LSTM and GRL show the best overall performance with 97% accuracy and very high F1-scores (0.96). CNN-LSTM achieves the highest recall (0.99), meaning it detects almost all seizures. GRL performs nearly as well (0.96 recall) while improving generalization through domain adaptation.

DA-SCT delivers balanced performance across all metrics, while SCT shows lower recall (0.79), indicating more missed seizures.

Overall, CNN-LSTM and GRL are the strongest models, with GRL offering better cross-patient robustness.

Future Implementation

AI-powered seizure prediction dashboard designed for hospitals and research environments. It integrates advanced deep learning models, including GRL-based domain adaptation, into a clean and scalable web interface.

Core Functionalities

Seizure Risk Forecast

- Real-time seizure probability display
- Time-to-onset estimation
- High-risk alerts for early intervention

Performance Summary

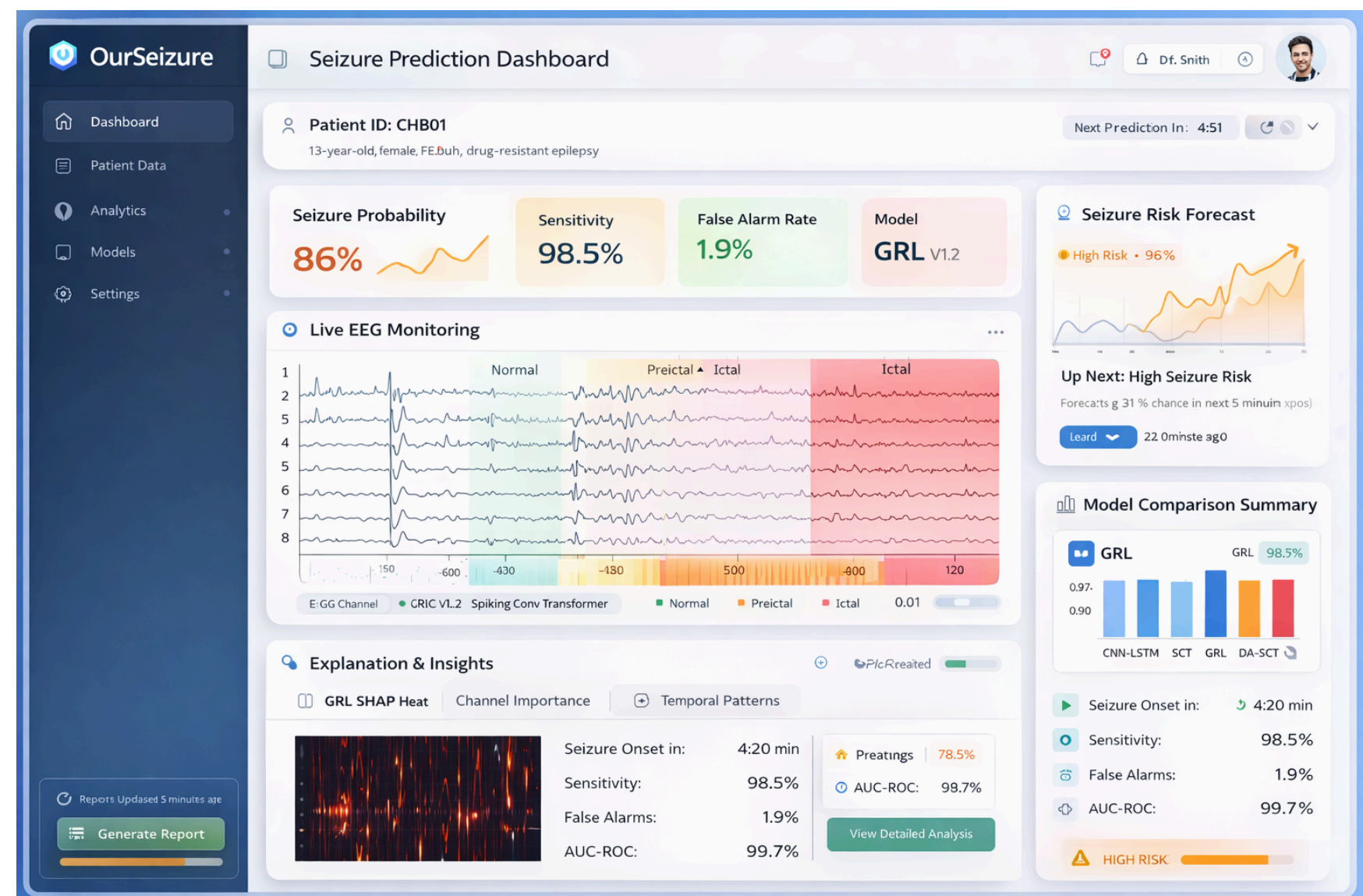
- Sensitivity: 98.5%
- False Alarm Rate: 1.9%
- AUC close to 0.99
- Confusion matrix and ROC tracking

Explainable AI Module

- SHAP heatmaps for channel-wise importance
- Temporal pattern visualization
- Transparent decision support for clinicians

Model Comparison

- CNN-LSTM
- Spiking Convolutional Transformer (SCT)
- GRL (Domain Adaptation)
- DA-SCT



XAI

THANK YOU

